

1. Confidence Interval (95%)

- Formula $CI = \bar{x} \pm (Z^* \times \frac{s}{\sqrt{n}})$

- what it measures: It gives an estimated range of values which likely to include an unknown population parameter (eg true average age or weight of the population) at a 95% confidence level.

- How to interpret: If the 95% CI age is $[42.2, 46.57]$, we can be 95% confident that the population true mean falls within this interval.

2. Hypothesis Testing: Two sample t-test / z-test

- Formula:
$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

what it measures: Compares the means of two groups of two independent groups (eg Male BMI vs Female BMI)

Page No. _____
Date: / /

4. One-way ANOVA (Analysis of Variance)

• Formula $F = \frac{\text{Mean Square Between Groups}}{\text{Mean Square within groups}}$

• what it measures: Evaluates whether the mean of a continuous variable (e.g. BMI) differs significantly across three or more categorical groups (e.g. Age Groups: 18-25, 26-35, 36-45, 46-60, 60+)

• Decision Rule:

If $p\text{-value} < 0.05$: reject $H_0 \rightarrow$ At least one age group has significantly different average BMI

5. Covariance and Pearson Correlation

Covariance Formula $\text{Cov}(X, Y)$

$$= \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n-1}$$

Pearson Correlation (r) =

$$\frac{\text{Cov}(X, Y)}{s_x \cdot s_y}$$

to see if they are statistically different from each other.

• Decision Rule:

- If p -value < 0.05 : Reject the Null Hypothesis (H_0) $\rightarrow$ There is a statistically significant difference between the two group means.

- If p -value > 0.05 : fail to reject $H_0 \rightarrow$ No significant difference.

5 Chi-square (χ^2) Test of Independence

- Formula $\chi^2 = \sum \frac{(O - E)^2}{E}$

(where O = observed frequency, E = Expected frequency under H_0)

- What it measures: Tests whether two categorical variables (e.g. smoking status vs. Diabetes diagnosis) are associated or independent.

- If p -value < 0.05 reject $H_0 \rightarrow$ The two variables significantly associated (dependent).

- What measures?
- Covariance: Shows the direction of the linear relationship between two numerical variables (e.g. Age and BMI).
- Correlation (r): Standardizes covariance between -1 and +1 to show both direction and stem strength.
- $r > 0$: Positive linear relationship
- $r < 0$: Negative linear relationship
- $r \approx 0$: No linear relationship